Cache Design Exploration

Direct-Mapped vs 2-Way vs 4-Way Set Associative Cache

Noah Dumas

CS 4200: Computer Architecture

Instructor: Dr. Zanyar ZohourianShahzadi

University of Colorado Colorado Springs

Spring 2026

Simulator Design

The simulator processes load and store operations from trace files.

Cache Type	Size	Block Size	Associativity
Direct-Mapped	256 bytes	16 bytes	1-way
2-Way Set Associative	256 bytes	16 bytes	2-way
4-Way Set Associative	256 bytes	16 bytes	4-way

Each access decodes the address, checks the selected set, records hit or miss, applies LRU replacement if needed, and tracks dirty writebacks.

Important Code: Address Decoding

This function splits each memory address into the parts needed for cache lookup.

```
def decode_address(self, address):
    block_number = address // self.block_size_bytes
    block_offset = address % self.block_size_bytes
    set_index = block_number % self.num_sets
    tag = block_number // self.num_sets

    return tag, set_index, block_offset, block_number
```

The index chooses the cache set, the tag verifies the block, and the offset identifies the byte inside the block.

Important Code: Hit and Store Handling

This section checks whether the selected cache set contains the requested block.

```
for way, line in enumerate(cache_set):
    if line.valid and line.tag == tag:
        self.stats.hits += 1
        line.last_used = self.clock

        if operation == "S":
            line.dirty = True

        return "HIT"

self.stats.misses += 1
```

On a store hit, the line is marked dirty because memory is updated later under the write-back policy.

Important Code: LRU and Dirty Writeback

On a miss, the simulator chooses a victim line and checks whether dirty data must be written back.

```
victim_way = self._choose_victim_way(cache_set)
victim_line = cache_set[victim_way]

if victim_line.valid:
    self.stats.evictions += 1

    if victim_line.dirty:
        self.stats.dirty_writebacks += 1

victim_line.valid = True
victim_line.dirty = operation == "S"
victim_line.tag = tag
victim_line.last_used = self.clock
```

This is where replacement policy, evictions, and write-back behavior are handled.

Running the Cache Simulator

`cache_design_exploration.py`

Direct-Mapped vs 2-Way vs 4-Way